AI Speech and language variation & change

A study on manifestation of AI speech in human natural language

class: [16838] germanische sprachen im vergleich
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Preliminary

inspired by BSI (2025) and Yakura et al. (2025) which found evidence for GPT influenced human language after the introduction of chatGPT we tried to replicate the Yakura et al. (2025) pipeline of building an AI vocabulary (model preferred lemmata) and compare frequencies of model typical words across pre- and post model introduction human language corpora. The first draft essai proves their hypothesis that LLM generated language manifests within human natural language.

Questions / Hypotheses

- do humans adapt to language produced by LLM and incorporate model vocabulary into their own language production?
- H1: we will find higher frequencies of LLM typical vocabulary within human natural language corpora after onset of model introduction.

References

BSI, Brand Science Institute. 2025. "Wie KI Unsere Sprache Verändert – Eine Empirische Studie." https://www.bsi.ag/cases/104-case-studie-wie-ki-unsere-sprache-veraendert---eine-empirische-studie.html.

DIP. 2026. "DIP - Bundestagsprotokolle." Docs. *DIP - API*. Berlin. https://dip.bundestag.de/%C3%BCber-dip/hilfe/api#content.

Yakura, Hiromu, Ezequiel Lopez-Lopez, Levin Brinkmann, Ignacio Serna, Prateek Gupta, Ivan Soraperra, and Iyad Rahwan. 2025. "Empirical Evidence of Large Language Model's Influence on Human Spoken Communication." arXiv. https://doi.org/10.48550/arXiv.2409.01754.

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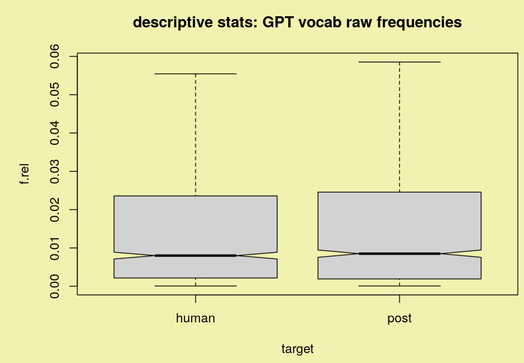

Figure 1: stats descriptive

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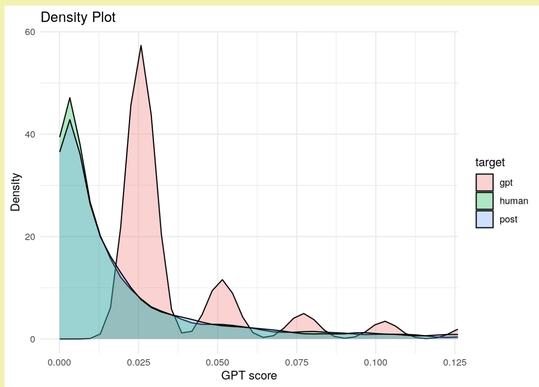

Figure 2: model vocab density

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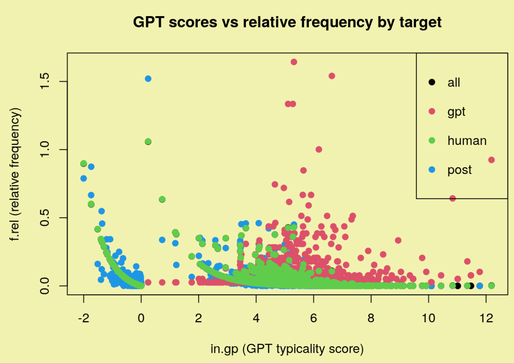

Figure 3: model vocab distribution

Methods / Data

our human languaga data consists of raw texts from german bundestag plenary protocols, DIP (2026). the LLM corpus consists of model summaries of a first subset of these texts generated with the prompt you see C1.

we first devised model typical lemmata in the AI corpus which are distinct for that corpus using a linear regression model that calculates a score for each lemma in the corpus, see Figure 3 and Figure 2.

```
C1: System prompt: C1: You are a member of german
parliament. Prepare a summary of the text
provided to present at a local community meeting
of your party members. Output in german language,
no preamble, no extra information, just the plain
text. Wordcount maximal 300 words, containing not
more than 5% of the keywords of the text provided
and explicitly not just a list of keywords but an
entertaining text. You are supposed to interprete
freely, including background insights on daily
politics. Keep in mind thatthe text will be used
as is as keynotes to the talk being held to the
locals. C1: Text:
```

```
corpus stats

|target |   freq|
|:------|------:|
|human  | 0.3708|
|post   | 0.3890|
|DIFF:  | 0.0182|
```

Results

to gather an insight, yet with simple descriptive stats comparing the raw frequencies of gpt-preferred lemmas in pre- and post-gemini onset we find that in the target corpus the occurences of these lemma increase, see Figure 1.

```
re tes
ja") #
s.c),
an language
  )
lm2a_glmer <- glmer(f.rel ~ target * in.gp + (1 | lemma),
##############################################################################
,
# with
  cmodel<-lapply(seq_along(1:ceiling(as.double(max(lpt$in.gp)))), function(i){
qdf$value[grep("key",qdf$key)]<-Sys.getenv("DIP_APIKEY")
lmdf.s<-lmdf.c[c(which(mh),which(mp)),]
ening/
summary texts
# legend("topright",
get.lemmas<-function(pt=0.2){
f1a1<-data.frame(f1a,target="all",size=sum(f1a$FREQ))
tok.r2[grepl("afd",tok.r2$word),]
#             lwd    = 2)
# for(i in seq_along(targets)) {
#tokens.r$lemma <- fj$lemma[match(tokens.r$word, fj$WORD)]
  family = poisson("sqrt"),
#lm2<-lmer(f.rel~target+in.gp+(1|lemma),lmdf.s)
#                   main = "Distribution of GPT scores (in.gp) by target",
### > more occurences (rel. frequencies) of gpt vocabular in post gpt corpus
# Tokenize to words (lowercase, remove punctuation/numbers)
# lm1<-lmer(FREQ~target*gu+(1|size),lmdf)
### get responsible lemmata
}
### get GPT score
m1 <- lmer(f.pm ~ target + (1 | WORD), data = lmdf.c)
#lm2b<-lmer(f.rel~target*gus.c+in.gp+(1|lemma),lmdf.s) ### or this, target pre/post
##############################################
# tu<-unique(tokens.2$word)
tok.r3<-tok.r3[!is.na(tok.r3$word),]

tdf2<-getdf(range)
t<-content(response,"text")
  coord_cartesian(xlim = c(0, 0.12)) +  # set your range here
# # we get n=9 lemma, freq from 303 to 1616
  mf<-ceiling(mf*100)
lpt[lpt$in.gp>=6&lpt$f.r2>=0.02,]
ptdf.2$target<-"post"
```